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Student Number

2015

TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Extension 1 Mathematics

21st July 2015

General Instructions

- Reading time – 5 minutes
- Working time - 2 hours
- Write using blue or black pen
Black pen is preferred
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11-14 show relevant mathematical reasoning and/or calculations
- Start a new booklet for each question

Total Marks – 70

Section I - Pages 2 – 5

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II - Pages 6 - 11

60 marks

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

Question	Marks
1 - 10	/10
11	/15
12	/15
13	/15
14	/15
Total	/70

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

This assessment task constitutes 40% of the Higher School Certificate Course Assessment

Section I

10 marks

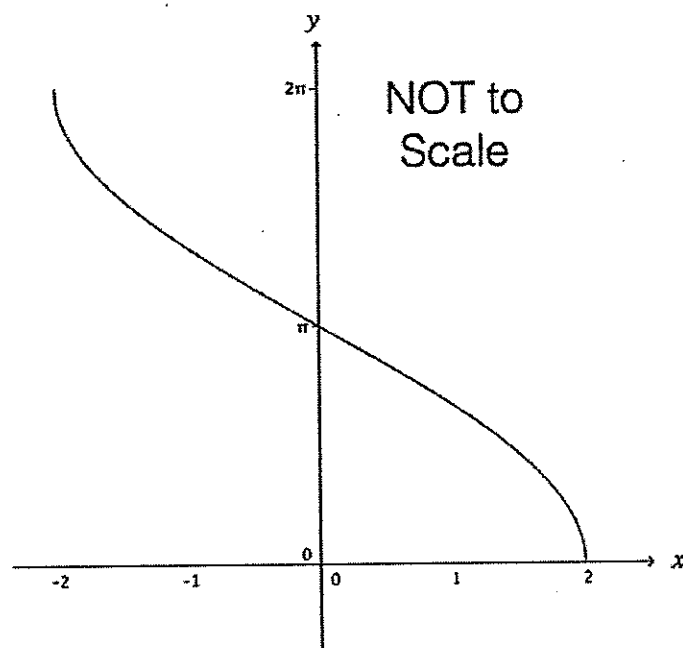
Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for questions 1 – 10 (Detach from paper)

- 1 If $(2k+1)x^2 - (k+2)x + 1$ is a perfect square then the value of k is:
- (A) 0
(B) 4
(C) 0 and 4
(D) -2
- 2 The point P divides the interval from $A(-2, 2)$ to $B(8, -3)$ internally in the ratio 3: 5.
What is the x -coordinate of P ?
- (A) $\frac{1}{2}$
(B) $\frac{7}{4}$
(C) $\frac{17}{4}$
(D) 3
- 3 The polynomial equation $3x^3 - 2x^2 + 3x - 4 = 0$ has roots α, β and γ . The exact value of $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma}$ is:
- (A) 2
(B) -2
(C) $\frac{1}{2}$
(D) $-\frac{1}{2}$

- 4 What is the equation of the graph below:



- (A) $y = 2 \sin^{-1}(2x)$
- (B) $y = 2 \sin^{-1}\left(\frac{x}{2}\right)$
- (C) $y = 2 \cos^{-1}(2x)$
- (D) $y = 2 \cos^{-1}\left(\frac{x}{2}\right)$
- 5 In how many ways can 6 different keys be arranged on a circular key ring?
- (A) 720
- (B) 120
- (C) 60
- (D) 6

6 The sum of the infinite geometric series $1 + 2^n + 2^{2n} + \dots$ is 2. The value of n is:

- (A) 3
- (B) $\frac{1}{2}$
- (C) -1
- (D) 2

7 Which expression is equivalent to $\int \cos^2 4x \, dx$?

- (A) $\frac{1}{2} \left(\frac{1}{8} \sin 8x - x \right) + C$
- (B) $\frac{1}{2} \left(\frac{1}{8} \sin 8x + x \right) + C$
- (C) $\frac{1}{2} \left(\frac{1}{4} \sin 4x - x \right) + C$
- (D) $\frac{1}{2} \left(\frac{1}{4} \sin 4x + x \right) + C$

8 When a polynomial is divided by $(x-2)(x+4)$ the remainder is $2x-5$.

What is the remainder when it is divided by $x-2$?

- (A) -5
- (B) -1
- (C) 1
- (D) 5

9 What is the derivative of $\sin^{-1}(2x)$

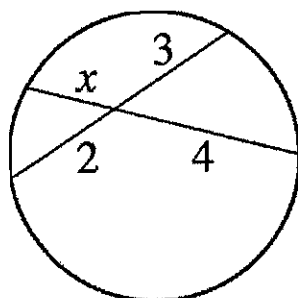
(A) $\frac{1}{2\sqrt{1-4x^2}}$

(B) $\frac{-1}{2\sqrt{1-4x^2}}$

(C) $\frac{2}{\sqrt{1-4x^2}}$

(D) $\frac{-2}{\sqrt{1-4x^2}}$

10



The value of x in the above diagram is:

(A) $\frac{3}{2}$

(B) $\frac{8}{3}$

(C) 6

(D) $\frac{2}{3}$

Section II

70 marks

Attempt Questions 11 – 14

Allow about 1 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

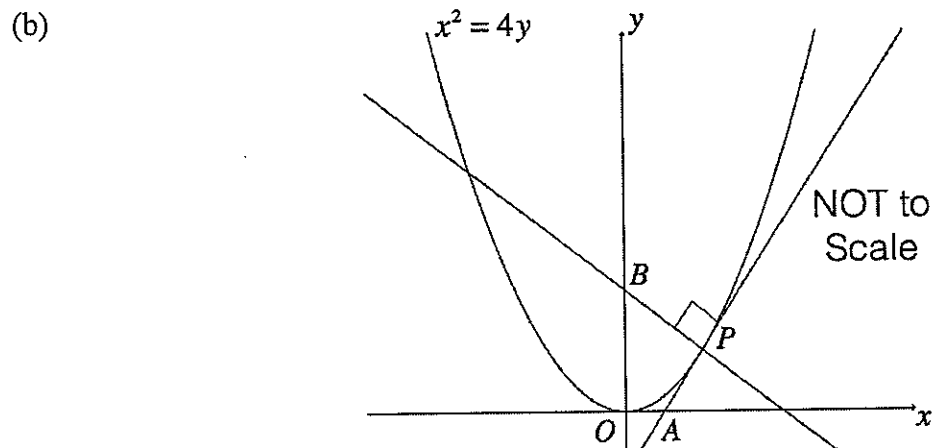
Question 11(15 marks) Use a SEPARATE writing booklet

- (a) Evaluate $\int_0^2 \frac{4}{\sqrt{4-x^2}} dx$ 3
- (b) Differentiate $\cos^{-1}(\sin x)$ and fully simplify. 2
- (c) Solve $x-3 < \frac{4}{x}$ 3
- (d) Use the substitution $x = \sin t$ to evaluate $\int_0^{\frac{1}{2}} \sqrt{1-x^2} dx$ 3
- (e) The parabola's $y = x^2$ and $y = (x-2)^2$ intersect at the point P .
- (i) Find the coordinates of P 1
- (ii) Find the angle between the tangents to the parabolas at P . Give your answer to the nearest degree. 3

End of Question 11

Question 12(15 marks) Use a SEPARATE writing booklet

- (a) Use mathematical induction to prove that $9^{n+2} - 4^n$ is divisible by 5 for integers $n \geq 1$. 3



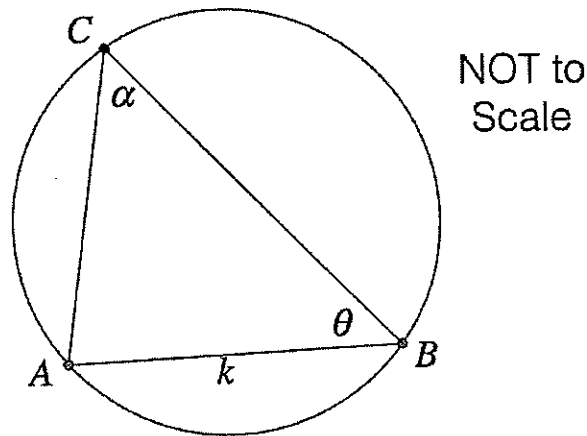
The diagram shows the graph of $x^2 = 4y$. The tangent to the parabola at $P(2p, p^2)$, where $p > 0$ cuts the x -axis at A . The normal to the parabola at P cuts the y -axis at B .

- (i) Find the equation of the tangent at P . 2
- (ii) Show that B has coordinates $(0, p^2 + 2)$ 2
- (iii) Let C be the midpoint of AB . Find the equation of the locus of C . 3
- (c) A particle P moving in a straight line executes Simple Harmonic Motion about a centre O . The acceleration of P is given by $\ddot{x} = -n^2x$ where x is the distance OP and n is a constant. The amplitude of the simple harmonic motion is a .
- (i) Show that $v^2 = n^2(a^2 - x^2)$ 3
- (ii) Show that the motion of the particle is described by $x = a \sin(nt + \alpha)$ where α is a constant 2

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet

(a)



Points A , B , and C lie on the circle

The length of the chord AB is a constant, k . The radian measures of $\angle ABC$ and $\angle BCA$ are θ and α respectively.

- (i) Let l be the sum of the length of the chord CA and CB . Show that l is given by: 3

$$l = \frac{k}{\sin \alpha} (\sin \theta + \sin(\theta + \alpha))$$

- (ii) Explain why is α a constant? 1

- (iii) Evaluate $\frac{dl}{d\theta}$ when $\theta = \frac{\pi}{2} - \frac{\alpha}{2}$ 2

- (iv) Hence show that l is a maximum when $\theta = \frac{\pi}{2} - \frac{\alpha}{2}$ 1

- (b) Sketch $y = \frac{x^2 + x - 2}{x^2 - 1}$ clearly showing all important features. 3

Question 13 continues on page 9

Question 13 (continued)

- (c) Let T be the temperature of an object at time t and D be the temperature of the surrounding medium. Newton's Law of Cooling states that the rate of change of T is proportional to $T - D$

$$\text{i.e. } \frac{dT}{dt} = -k(T - D)$$

- (i) Show that $T = D + Ce^{-kt}$ (where C and k are constants) satisfies Newton's Law of cooling

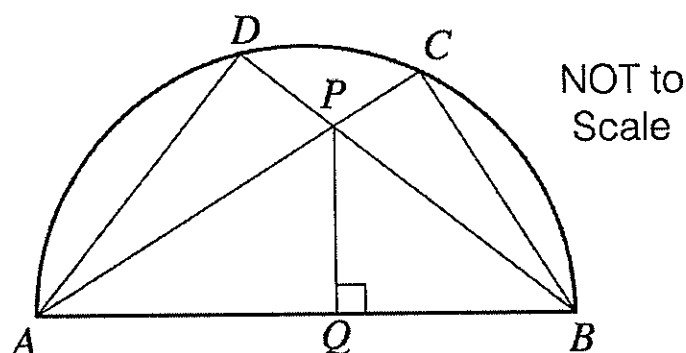
1

- (ii) A packet of meat with initial temperature of 25°C is placed in the freezer whose temperature is kept at a constant -10°C . It takes 12 minutes for the temperature of the meat to drop to 15°C . How much *additional* time is needed for the temperature of the meat to fall to 0°C . Give your answer in minutes correct to the nearest second.

4

End of Question 13

Question 14(15 marks) Use a SEPARATE writing booklet



(a) AB is the diameter of a semi-circle and $PQ \perp AB$

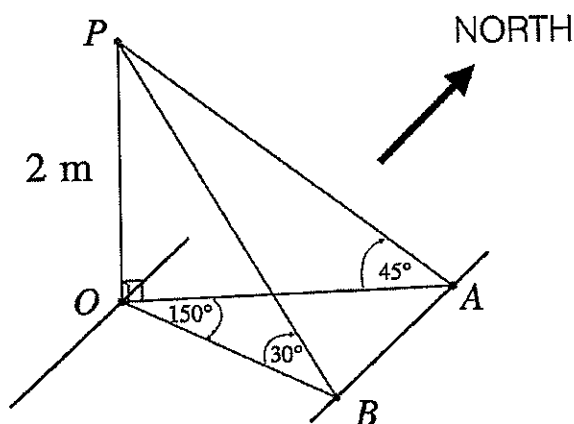
(i) Explain why $AQPD$ and $BQPC$ are cyclic quadrilaterals

1

(ii) Hence or otherwise prove that PQ bisects $\angle DQC$

3

(b) A vertical pole of height 2 m with base at point O stands on the West side of a canal that has straight parallel sides running from North to South. Two points A and B both lie on the East side of the canal, A to the north of the pole and B to the south of the pole, such that $\angle AOB = 150^\circ$. The angles of elevation of the top of P of the pole. The angles of elevation to the top P of the pole are 45° from A and 30° from B .



Find:

(i) The exact distance AB

2

(ii) The width of the canal (the perpendicular distance between opposite banks of the canal). Answer to the nearest centimetre.

3

Question 14 continues on page 11

Question 14 (continued)

- (c) A particle is projected from a point O on the ground with a velocity V and an inclination of θ to the horizontal. After time t the horizontal distances travelled by P are

$$x = Vt \cos \theta \text{ and } y = Vt \sin \theta - \frac{gt^2}{2} \text{ respectively. DO NOT PROVE THIS.}$$

- (i) If $\tan \theta = \frac{1}{3}$ and P passes through the point $A \left(3a, \frac{3a}{4} \right)$ show that 3
- $$V^2 = 20ga$$

- (ii) At the instant of time when P is travelling in the horizontal direction (i.e. it has no vertical component to its motion), another particle Q is projected from O with a velocity U at an angle α to the horizontal. P and Q hit the ground at the same place and at the same time. 3

$$\text{Show that } U = \sqrt{\frac{145ga}{2}} \text{ and that } \alpha = \tan^{-1} \left(\frac{1}{12} \right)$$

End of Examination ☺

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Mathematics

Section I – Multiple Choice Answer Sheet

Use this multiple-choice answer sheet for questions 1 – 10. Detach this sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct

Start
Here



1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

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Ex1 Mathematics Trial 2015 Solution

(1) C (2) B (3) C (4) D (5) C (6) C (7) B (8) B (9) C (10) A

Question 11(15 marks)

(a)

$$\int_0^2 \frac{4}{\sqrt{4-x^2}} dx = 4 \left[\sin^{-1} \frac{x}{2} \right]_0^2$$

$$= 4(\sin^{-1} 1 - 0) = 4 \times \frac{\pi}{2}$$

$$= 2\pi$$

3 marks obtains a correct solution

2 marks obtains correct primitive

1 mark identifies inverse trig attempts solution

(b)

$$\frac{d}{dx} \cos^{-1}(\sin x) = \frac{-\cos x}{\sqrt{1-\sin^2 x}} = \frac{-\cos x}{\cos x} = -1$$

1 correct differentiation of $\cos^{-1}$

1 correct simplified answer

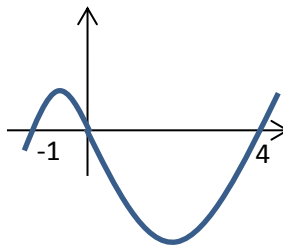
(c)

$$x^3 < \frac{4}{x} \quad x \neq 0$$

$$x(x^2 - 3x - 4) < 0$$

$$x(x-4)(x+1) < 0$$

$$\therefore x < -1, \quad 0 < x < 4$$



1 obtaining the cubic function

2 marks obtains one correct interval, r equivalent

3 marks correct solution

(d)

$$x = \sin t$$

$$dx = \cos t \, dt$$

$$\sqrt{1-x^2} = \sqrt{1-\sin^2 t}$$

$$= \cos t$$

$$\text{When } x = 0, t = 0$$

$$x = \frac{1}{2}, t = \frac{\pi}{6}$$

$$\frac{1}{2} \int_0^{\frac{\pi}{6}} \sqrt{1-x^2} \, dx = \int_0^{\frac{\pi}{6}} \cos^2 t \, dt$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{6}} \cos 2t + 1 \, dt$$

$$= \frac{1}{2} \left[\frac{\sin 2t}{2} + t \right]_0^{\frac{\pi}{6}}$$

1 finding all the correct substitutions including the integrand.

1 correctly integrate

1 correct answer

$$= \frac{1}{2} \left(\frac{\sqrt{3}}{4} + \frac{\pi}{6} \right) = \frac{\sqrt{3}}{8} + \frac{\pi}{12}$$

(e)

$$y = x^2 \text{ and } y = (x - 2)^2$$

$$x^2 = (x - 2)^2$$

$$4x - 4 = 0$$

$$x = 1, y = 1$$

$$P(1, 1)$$

$$\frac{d}{dx} x^2 = 2x$$

$$\text{At } x = 1$$

$$m_1 = 2$$

$$\frac{d}{dx} (x - 2)^2 = 2x - 4$$

$$m_2 = -2$$

$$\theta = \tan^{-1} \left| \frac{2 + 2}{1 + 2 \times -2} \right|$$

$$= 53^\circ$$

1 correct coordinates of P

1 correct gradients

1 correct sub. into formula

1 marks answer correct to nearest degree

($\frac{0}{2}$ for using incorrect formula)

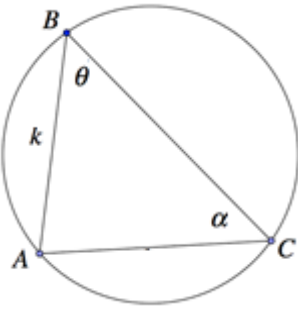
Question 12 (15 marks)

(a) $9^{n+2} - 4^n$ Prove it is true for $n = 1$ $9^{1+2} - 4^1 = 9^3 - 4$ $= 725 \text{ is divisible by } 5$ $\therefore$ It is true for $n = 1$ Assume it is true for $n = k$ ie. $9^{k+2} - 4^k = 5P$, where P is an integer R.T.P it is true for $n = k+1$ ie. $9^{k+3} - 4^{k+1} = 5Q$, where Q is an integer $\begin{aligned} 9 \cdot 9^{k+2} - 4^{k+1} &= 9^{k+2}(5 + 4) - 4^{k+1} \\ &= 5 \cdot 9^{k+2} + 4 \cdot 9^{k+2} - 4^{k+1} \\ &= 5 \cdot 9^{k+2} + 4(9^{k+2} - 4^k) \text{ by assumption} \\ &= 5 \cdot 9^{k+2} + 4(5P) \\ &= 5(9^{k+2} + 4P) \\ &= 5Q \text{ where } Q = 9^{k+2} + 4P \end{aligned}$ $\therefore$ Statement is true for $n = k+1$, $\therefore$ Statement is true for all $n \geq 1$ by mathematical induction.	1 proving $n = 1$ 1 assumption and R.Q.T $9^{k+2} - 4^k = 5P$ 1 correct process to obtain the result.
(b) (i) $y = \frac{x^2}{4}$ $\frac{dy}{dx} = \frac{x}{2}$ At $x = 2p$, $m = p$ Equation of the tangent is $y - p^2 = p(x - 2p)$ $px - y - p^2 = 0$ (ii) gradient of the normal is $-\frac{1}{p}$ equation of the normal is $y - p^2 = -\frac{1}{p}(x - 2p)$ $py + x - p^3 - 2p = 0$ when $x = 0$, $y = p^2 + 2$ $\therefore$ B has coordinates $(0, p^2 + 2)$ (iii) When $y = 0$, $x = p$ $\therefore$ A has coordinates $(p, 0)$ The midpoint of AB is	1 finding gradient 1 correct equation of tangent at P with full working 1 correct equation of normal 1 correct coordinates of B 1 correct coordinates of A 1 correct mid-point of AB

$M\left(\frac{p+0}{2}, \frac{0+p^2+2}{2}\right)$ $x = \frac{p}{2} \text{ and } y = \frac{p^2+2}{2}$ $p = 2x \text{ sub. into } y$ $y = \frac{4x^2+2}{2}$ $\therefore y = 2x^2 + 1 \text{ is the locus of P.}$ (c) (i) $\ddot{x} = -n^2x$ $\frac{d}{dx} \frac{1}{2} v^2 = -n^2x$ $\int \frac{d}{dv} \frac{1}{2} v^2 = - \int n^2 x dx$ $\frac{1}{2} v^2 = -\frac{1}{2} n^2 x^2 + C$ When $v = 0, x = a \therefore C = \frac{1}{2} n^2 a^2$ $\therefore v^2 = n^2 a^2 - n^2 x^2$ $= n^2 (a^2 - x^2)$ (ii) $x = a \sin(nt + \alpha)$ From (i) $v = \pm n \sqrt{a^2 - x^2}$ $\frac{dx}{dt} = \pm n \sqrt{a^2 - x^2}$ $n \frac{dt}{dx} = \frac{\pm 1}{\sqrt{a^2 - x^2}}$ $\int n dt = \int \frac{\pm 1}{\sqrt{a^2 - x^2}} dx$ $nt = \sin^{-1} \left(\frac{x}{a} \right) + C \text{ or } nt = \cos^{-1} \left(\frac{x}{a} \right) + C$ <i>We only need to consider positive value of v for the solution.</i> $\therefore nt + \alpha = \sin^{-1} \left(\frac{x}{a} \right) \text{ where } \alpha = -C$ $\frac{x}{a} = \sin(nt + \alpha)$ Hence $x = a \sin(nt + \alpha)$	1 correct Cartesian equation of locus of M 1 using correct formula of $\ddot{x}$ using $\frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ 1 correct integration 1 correct value of C 1 signification progress towards establishing the integral 1 correct integration and simplification to achieve desired result
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Question 13 (15 marks)

(a) (i)



$$l = CA + CB$$

$$\frac{CA}{\sin \theta} = \frac{k}{\sin \alpha}$$

$$CA = \frac{k \sin \theta}{\sin \alpha}$$

$$\frac{CB}{\sin(180^\circ - (\alpha + \theta))} = \frac{k}{\sin \alpha}$$

$$CB = \frac{k \sin(\alpha + \theta)}{\sin \alpha}$$

$$CA + CB = \frac{k \sin \theta}{\sin \alpha} + \frac{k \sin(\alpha + \theta)}{\sin \alpha}$$

$$\therefore l = \frac{k}{\sin \alpha} (\sin \theta + \sin(\alpha + \theta))$$

(ii)

α is the angle opposite the side AB which is a constant k . Therefore α is a constant.

(iii)

$$l = \frac{k}{\sin \alpha} (\sin \theta + \sin(\alpha + \theta))$$

$$\frac{dl}{d\theta} = \frac{k}{\sin \alpha} (\cos \theta + \cos(\theta + \alpha))$$

$$\text{Sub. } \theta = \frac{\pi}{2} - \frac{\alpha}{2}$$

$$\Rightarrow \frac{dl}{d\theta} = \frac{k}{\sin \alpha} \left(\cos\left(\frac{\pi}{2} - \frac{\alpha}{2}\right) + \cos\left(\frac{\pi}{2} - \frac{\alpha}{2} + \alpha\right) \right)$$

$$= \frac{k}{\sin \alpha} \left(\sin \frac{\alpha}{2} + \cos\left(\frac{\pi}{2} + \frac{\alpha}{2}\right) \right)$$

$$= \frac{k}{\sin \alpha} \left(\sin \frac{\alpha}{2} - \sin \frac{\alpha}{2} \right) = 0$$

(iii)

1 correct length of CA with working

1 correct length of CB with working

1 correct length of l

1 correct reason

1 correct derivative of l

1 showing $\frac{dl}{d\theta} = 0$

$$\begin{aligned}\frac{d^2l}{d\theta^2} &= \frac{k}{\sin \alpha} (-\sin \theta - \sin(\theta + \alpha)) \\ &= -\frac{k}{\sin \alpha} (\sin \theta + \sin(\theta + \alpha)) \\ &< 0\end{aligned}$$

Also $\sin \theta > 0$ and $\sin(\theta + \alpha) > 0$, as θ is acute and $\theta + \alpha < 180^\circ$

$$\frac{dl}{d\theta} = 0 \quad \text{when} \quad \theta = \frac{\pi}{2} \quad \frac{\pi}{2} \quad (\text{from ii})$$

$\therefore l$ is a maximum when $\theta = \frac{\pi}{2} \quad \frac{\pi}{2} \quad (\text{using (ii)})$

(c)

$$\begin{aligned}y &= \frac{x^2 + x - 2}{x^2 - 1} \\ &= \frac{(x+2)(x-1)}{(x+1)(x-1)} \\ &= \frac{x+2}{x+1}\end{aligned}$$

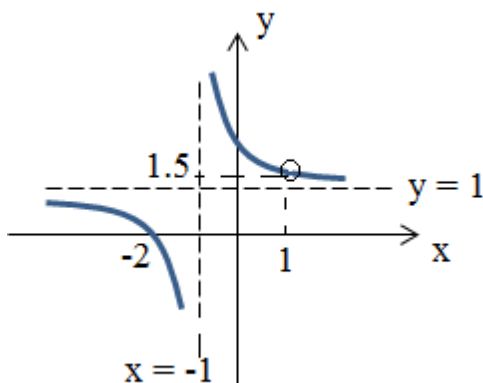
- Vertical asymptote $x = -1$
- Horizontal asymptote

$$y = \lim_{x \rightarrow \infty} \frac{x+2}{x+1} = 1$$

- The graph is discontinuous at $x = 1$ and

$$y = \lim_{x \rightarrow 1} \frac{x+2}{x+1} = \frac{3}{2}$$

- x -intercept is at $x = -2$
- y -intercept is at $y = 2$



(c)

(i)

$$T = D + Ce^{-kt}$$

$$\frac{dT}{dt} = -kCe^{-kt}$$

$$= -k(Ce^{-kt} + D - D)$$

$$= -k(T - D)$$

$\therefore$ the equation satisfies Newton's law of cooling.

1 showing $\frac{d^2l}{d\theta^2} < 0$ with conclusion

1 mark for correct asymptotes

1 mark for correct discontinuity

1 mark for correct shape

if x and y intercepts not labelled only subtract 1 mark

1 showing correct result

(ii)

$$T = -10 + Ae^{-kt}$$

$$\text{When } t = 0, T = 25 \Rightarrow 25 = -10 + A$$

$$\therefore A = 35$$

$$\text{When } t = 12\text{mins}, T = 15^\circ$$

$$15 = -10 + 35e^{-k \times 12}$$

$$k = \ln(25 \div 35) \div -12 \\ = 0.02804$$

$$\text{When } T = 0$$

$$0 = -10 + 35e^{-0.02804t}$$

$$t = \ln(10 \div 35) \div -0.02804$$

$$= 44.677709 \text{ mins}$$

$$= 44\text{mins and } 41 \text{ secs}$$

So it takes 32 minutes and 41 seconds additional time for the meat to drop to 0 degrees

1 correct value of A

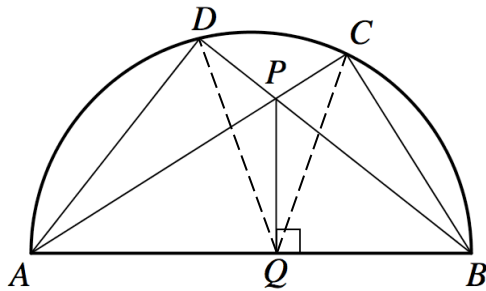
1 correct value of k

1 correct substitution into formula

1 correct answer (additional time)

Question 14 (15 marks)

(a)



(i)

$$\angle ADB = 90^\circ \text{ (angle in a semi-circle)}$$

$$\angle AQP = 90^\circ \text{ (given)}$$

$$\angle ADB + \angle AQP = 180^\circ$$

$\therefore$ ADPQ is a cyclic quadrilateral (sum of opposite angles is 180°)

Similarly QPCB is also a cyclic quadrilateral.

(ii)

$$\angle DAC = \angle CBD \text{ (angles in same segment subtended by arc DC)} \quad (1)$$

$$\angle DQP = \angle DAC \text{ (angles in same segment subtended by arc DP, using (i))}$$

$$\angle CQP = \angle CBD \text{ (angles in same segment subtended by arc PC, using (i))}$$

$$\therefore \angle DQP = \angle CQP \text{ (using (1))}$$

Hence QP bisects $\angle CQD$

1 showing sum of opposite angles is 180° with reason.

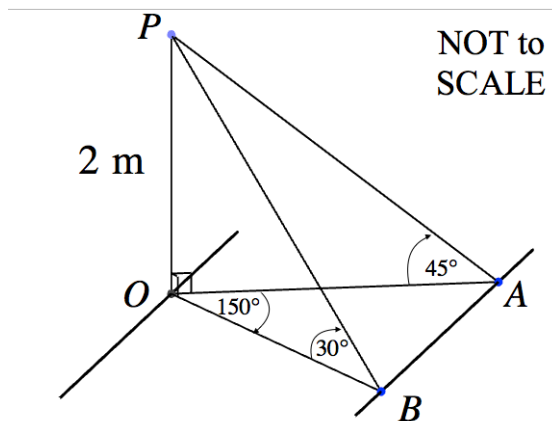
1 showing $\angle DAC = \angle CBD$ with reason

1 showing $\angle DQP = \angle DAC$ with reason

1 showing $\angle CQP = \angle CBD$ with reason

-1 if no conclusion

(b)



(i)

$$OB = \frac{2}{\tan 30} = 2\sqrt{3} \text{ m}$$

$$OA = \frac{2}{\tan 45} = 2 \text{ m}$$

$$AB^2 = (2\sqrt{3})^2 + 2^2 - 2 \times 2 \times 2\sqrt{3} \times 1 \times \cos 150^\circ$$

$$= 28$$

$$AB = \sqrt{28}$$

$$= 5.29 \text{ m}$$

1 correct values of OA & OB

1 correct value of AB

(ii)

$$\frac{\sin 150}{\sqrt{26}} = \frac{\sin \angle OAB}{2\sqrt{3}}$$

$$\sin \angle OAB = \frac{2\sqrt{3} \sin 150}{\sqrt{28}}$$

$$h = 2 \times \sin \angle OAB$$

$$= 2 \times \left(\frac{2\sqrt{3} \sin 150}{\sqrt{28}} \right)$$

$$= 0.65m$$

is the width of the cannal.

1 correct use of sin rule to find $\sin \angle OAB$

1 mark finding h

1 correct value of h to nearest cm

(c)

(i)

$$x = Vt \cos \theta \quad \text{and} \quad y = Vt \sin \theta - \frac{gt^2}{2}$$

$$t = \frac{x}{V \cos \theta} \quad \text{sub. into } y$$

$$y = x \tan \theta - \frac{g}{2} \left(\frac{x}{V \cos \theta} \right)^2$$

$$= x \tan \theta - \frac{gx^2}{2V^2} (\tan^2 \theta + 1) \quad (1)$$

$$\text{Sub. } \tan \theta = \frac{1}{3} \text{ and } A \left(3a, \frac{3a}{4} \right) \text{ into (1)}$$

$$\frac{3a}{4} = 3a \times \frac{1}{3} - \frac{9a^2 g}{2V^2} \left(\frac{1}{3^2} + 1 \right)$$

$$27aV^2 = 36aV^2 - 180a^2 g$$

$$V^2 = \frac{180a^2 g}{9a} = 20ag$$

1 finding Cartesian equation of the flight path of P

1 correct substitution

1 showing correct result

(ii)

At the instant when P is travelling in the horizontal direction (i.e. it has no vertical component to its motion),

this happens when P reaches the highest point, i.e. when $v_y = 0$, where $v_y = V \sin \theta - gt$

The time for P to reach the highest point is:

$$0 = V \sin \theta - gt$$

$$\therefore t = \frac{V \sin \theta}{g}$$

$$\text{Sub. } \sin \theta = \frac{1}{\sqrt{10}} \text{ and } V = \sqrt{20ga}$$

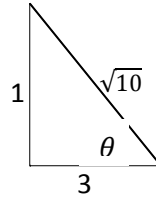
$\Rightarrow t = \frac{1}{g} \sqrt{\frac{20ga}{10}} = \sqrt{\frac{2a}{g}}$ is the time for the 2nd particle to reach max. range.

Hence the time for P to reach the maximum range is

$$t = 2\sqrt{\frac{2a}{g}}$$

And the maximum range for P is

$$\begin{aligned} x &= Vt \cos \theta \\ &= \sqrt{20ga} \times 2\sqrt{\frac{2a}{g}} \times \frac{3}{\sqrt{10}} \\ &= 12a \end{aligned}$$



The time for the second particle to reach the point

(12a, 0) is $t = \sqrt{\frac{2a}{g}}$

Sub. $x = 12a$ and $t = \sqrt{\frac{2a}{g}}$ into $x = Ut \cos \alpha$

$$\begin{aligned} U \cos \alpha &= \frac{12a}{U} \times \sqrt{\frac{g}{2a}} \\ &= \sqrt{72ga} \quad (2) \end{aligned}$$

Sub. $y = 0$ and $t = \sqrt{\frac{2a}{g}}$ into $y = Ut \sin \alpha - \frac{1}{2}gt^2$

$$0 = U \sin \alpha \times \sqrt{\frac{2a}{g}} - \frac{g}{2} \times \frac{2a}{g}$$

$$\Rightarrow U \sin \alpha = a \times \sqrt{\frac{g}{2a}} = \sqrt{\frac{ag}{2}} \quad (3)$$

$$(2)^2 + (3)^2$$

$$\begin{aligned} \Rightarrow U^2(\cos^2 \alpha + \sin^2 \alpha) &= 72ga + \frac{ag}{2} \\ &= \frac{145ga}{2} \end{aligned}$$

$$\therefore U = \sqrt{\frac{145ga}{2}}$$

$$(2)^2 \div (3)^2$$

1 finding the max. range for 2nd particle

1 simplifying algebraic expressions to obtain correct answer for U

1 simplifying algebraic expressions to obtain correct answer for α

$\frac{U \sin \alpha}{U \cos \alpha} = \sqrt{\frac{ag}{2}} \div \sqrt{72ga}$ $\Rightarrow \quad = \sqrt{\frac{ag}{144ag}}$ $\therefore \tan \alpha = \frac{1}{12}$ $\text{hence } \alpha = \tan^{-1}\left(\frac{1}{12}\right)$	
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